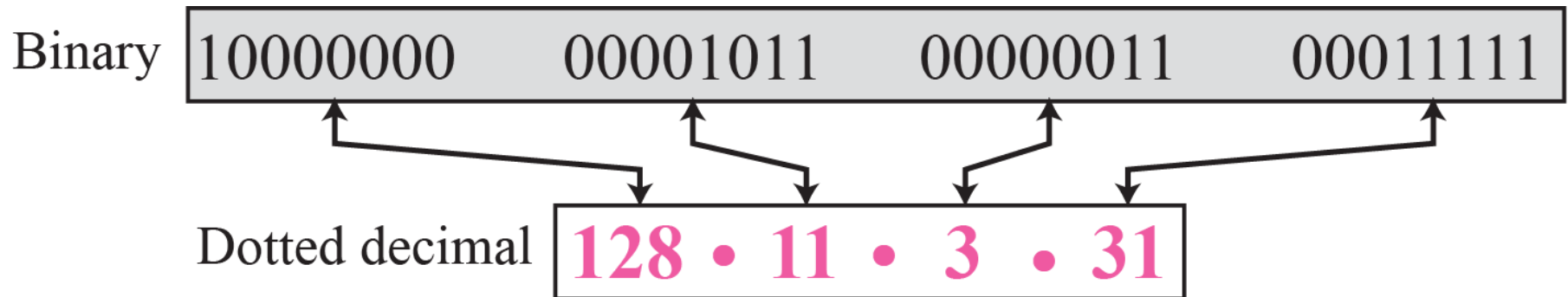


EETS 7302 TCP/IP Network Administration

Session 4 IPv4 Addressing

Dotted-decimal notation of IPv4 address



Example (5 min. exercise)

Change the following IPv4 addresses from binary notation to dotted-decimal notation.

a. 10000001 00001011 00001011 11101111

b. 11000001 10000011 00011011 11111111

Solution

?

Example (5 min. exercise)

Change the following IPv4 addresses from binary notation to dotted-decimal notation.

a. 10000001 00001011 00001011 11101111

b. 11000001 10000011 00011011 11111111

Solution

We replace each group of 8 bits with its equivalent decimal number (see Appendix B) and add dots for separation:

a. 129.11.11.239

b. 193.131.27.255

Example (5 min. exercise)

Find the error, if any, in the following IPv4 addresses:

a. 111.56.045.78

b. 221.34.7.8.20

c. 75.45.301.14

d. 11100010.23.14.67

Example (5 min. exercise)

Find the error, if any, in the following IPv4 addresses:

- a.** 111.56.045.78
- b.** 221.34.7.8.20
- c.** 75.45.301.14
- d.** 11100010.23.14.67

Solution

- a.** There should be no leading zeroes (045).
- b.** We may not have more than 4 bytes in an IPv4 address.
- c.** Each byte should be less than or equal to 255.
- d.** A mixture of binary notation and dotted-decimal notation.

Example (5 min. exercise)

Find the number of addresses in a range if the first address is 146.102.29.0 and the last address is 146.102.32.255.

Example (5 min. exercise)

Find the number of addresses in a range if the first address is 146.102.29.0 and the last address is 146.102.32.255.

Solution

We can subtract the first address from the last address in base 256. The result is 0.0.3.255 in this base. To find the number of addresses in the range (in decimal), we convert this number to base 10 and add 1 to the result..

$$\text{Number of addresses} = (0 \times 256^3 + 0 \times 256^2 + 3 \times 256^1 + 255 \times 256^0) + 1 = 1024$$

00011101 → 29

00011110 → 30

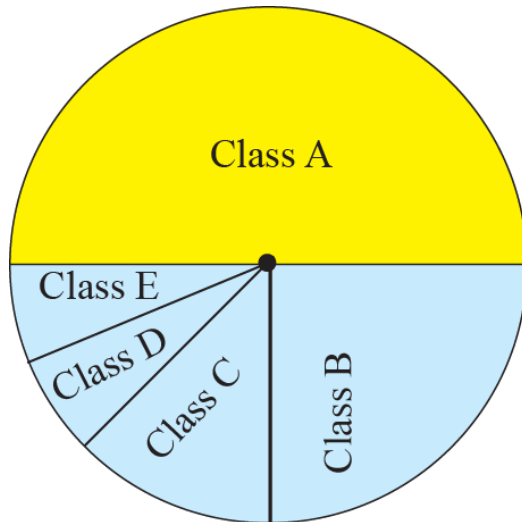
00011111 → 31

00100000 → 32

CLASSFUL ADDRESSING

- **IP addresses, when started a few decades ago, used the concept of classes.**
- **In the mid-1990s, a new architecture, called classless addressing, was introduced**

Occupation of address space



Class A: $2^{31} = 2,147,483,648$ addresses, 50%

Class B: $2^{30} = 1,073,741,824$ addresses, 25%

Class C: $2^{29} = 536,870,912$ addresses, 12.5%

Class D: $2^{28} = 268,435,456$ addresses, 6.25%

Class E: $2^{28} = 268,435,456$ addresses, 6.25%

Finding the class of address

	Octet 1	Octet 2	Octet 3	Octet 4
Class A	0.....			
Class B	10.....			
Class C	110.....			
Class D	1110....			
Class E	1111....			

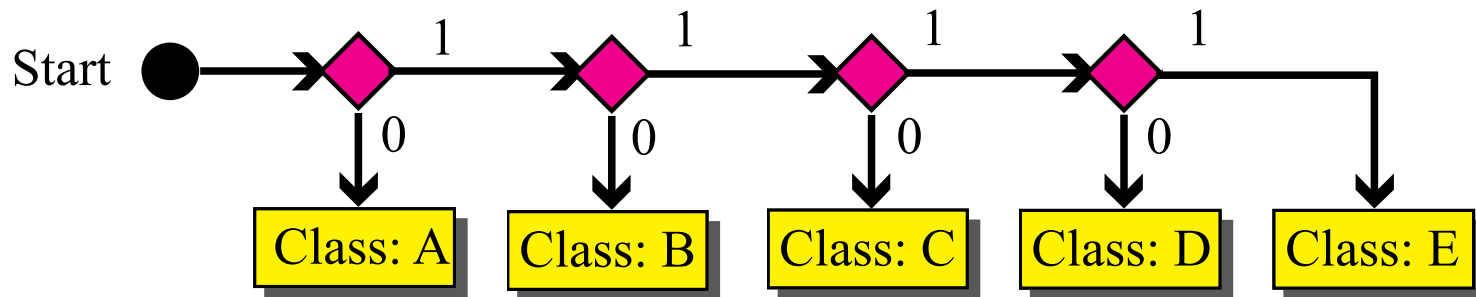
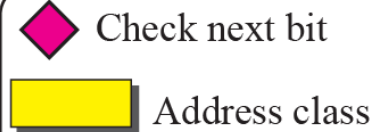
Binary notation

	Byte 1	Byte 2	Byte 3	Byte 4
Class A	0–127			
Class B	128–191			
Class C	192–223			
Class D	224–299			
Class E	240–255			

Dotted-decimal notation

Procedure for finding the class of an address

Legend



Example (5.min exercise)

Find the class of each address:

- a.** 00000001 00001011 00001011 11101111
- b.** 11000001 10000011 00011011 11111111
- c.** 10100111 11011011 10001011 01101111
- d.** 11110011 10011011 11111011 00001111

Example (5.min exercise)

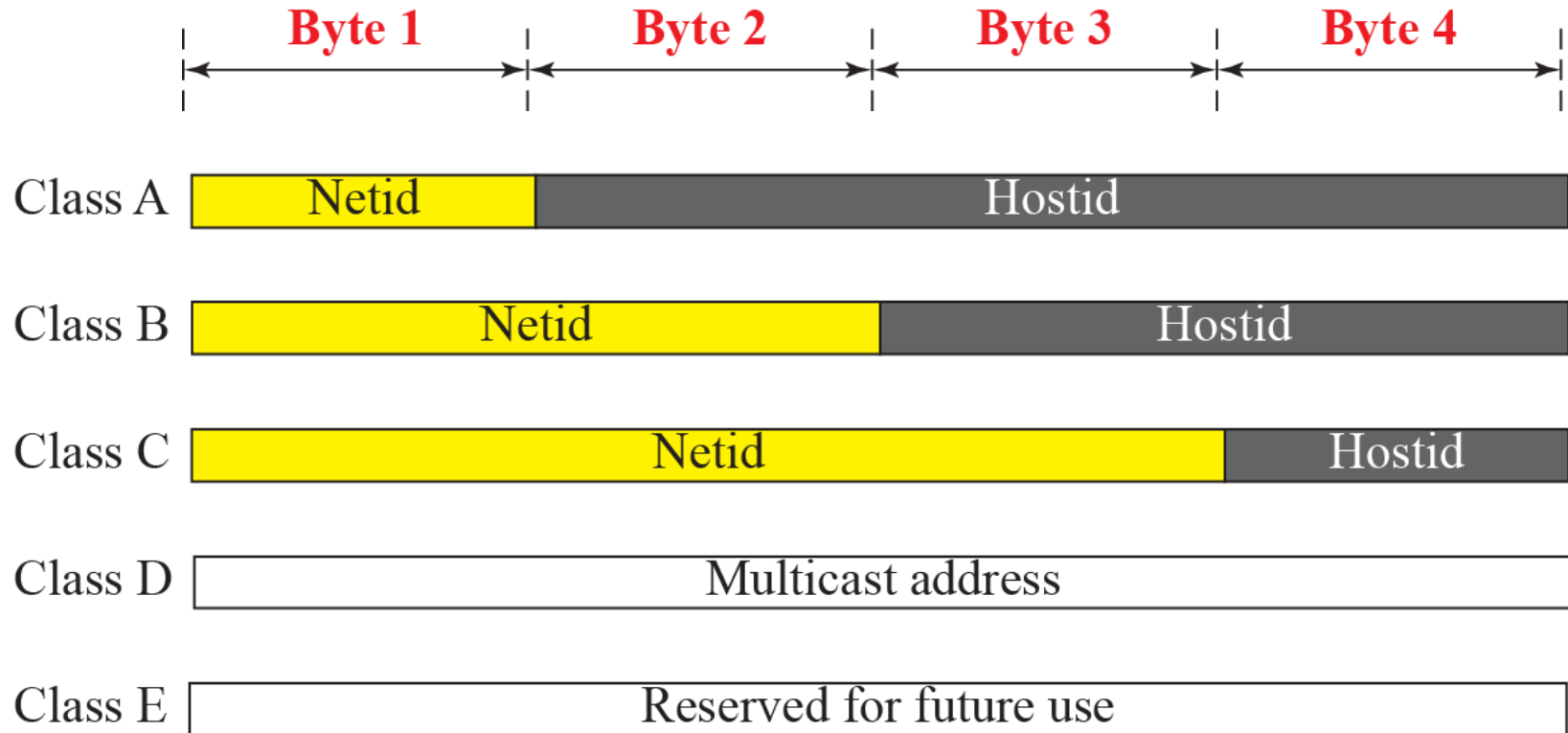
Find the class of each address:

- a.** 00000001 00001011 00001011 11101111
- b.** 11000001 10000011 00011011 11111111
- c.** 10100111 11011011 10001011 01101111
- d.** 11110011 10011011 11111011 00001111

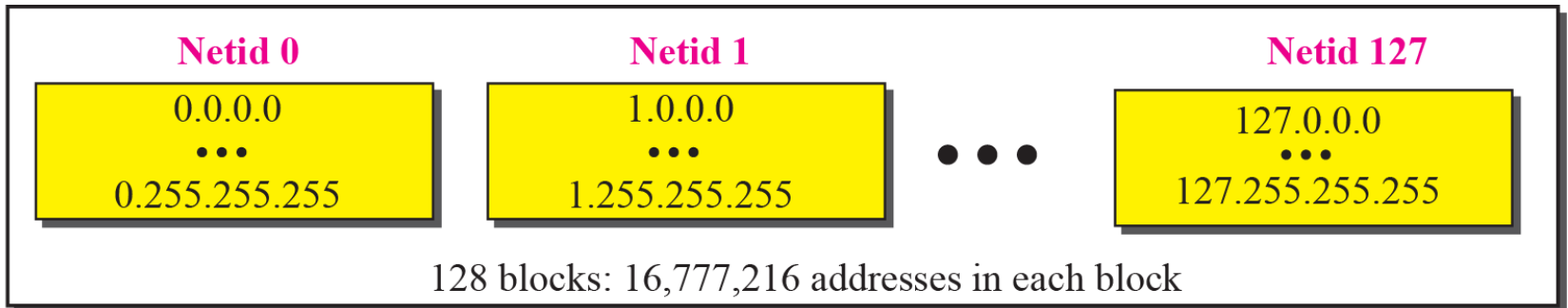
Solution

Use the address finding procedure

- a.** The first bit is 0. This is a class A address.
- b.** The first 2 bits are 1; the third bit is 0. This is a class C address.
- c.** The first bit is 1; the second bit is 0. This is a class B address.
- d.** The first 4 bits are 1s. This is a class E address.

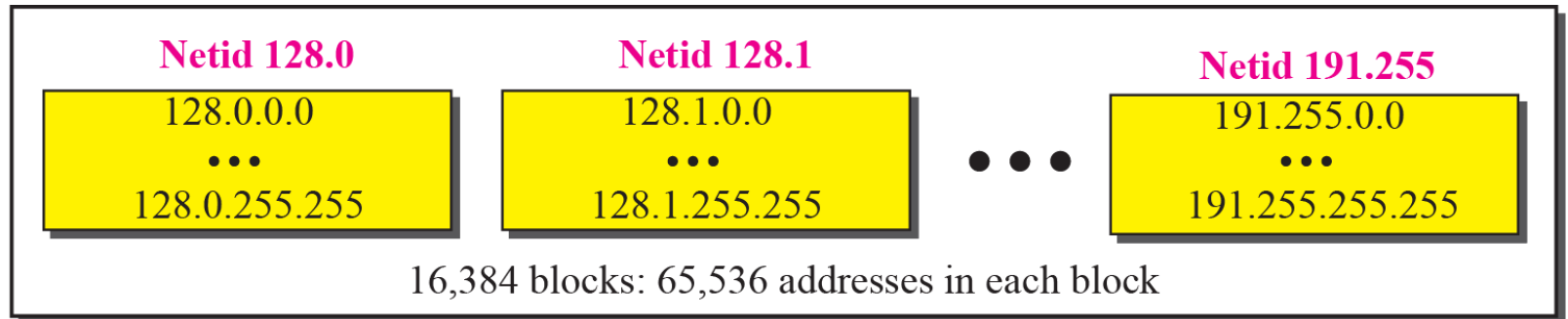


Class A

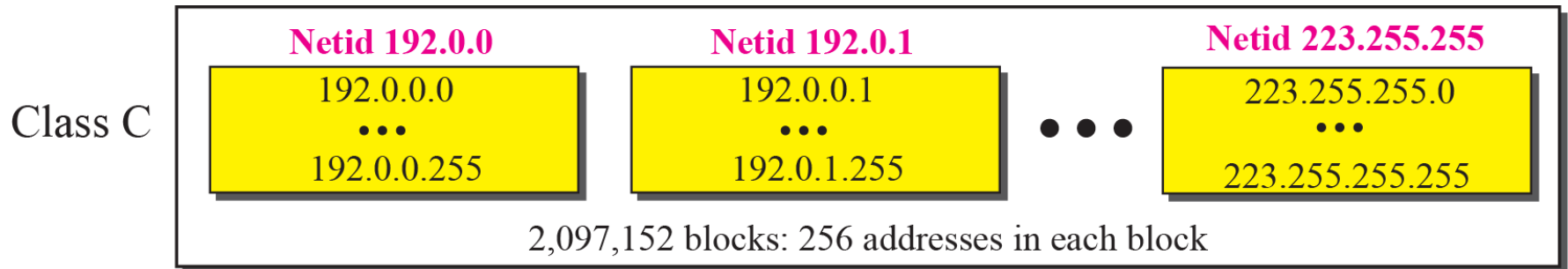


***Millions of class A addresses are wasted.
Organizations may not be that big..***

Class B



Many class B addresses are wasted.

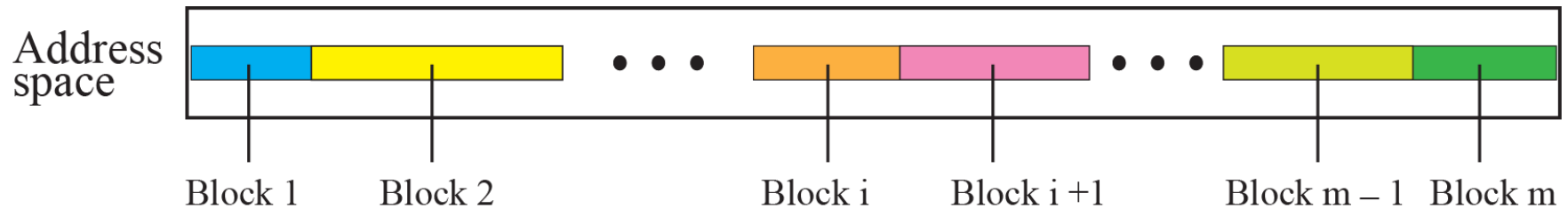


***Not so many organizations are so small to have a class C block.
Therefore, another addressing mechanism has been proposed..***

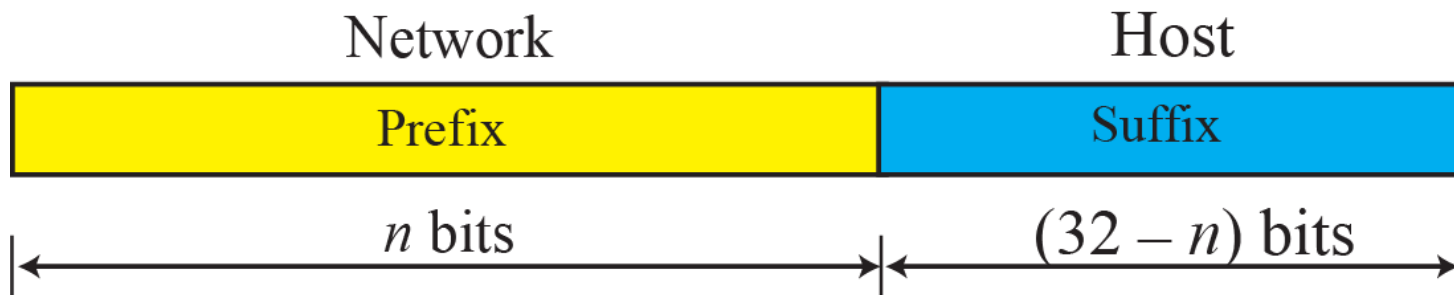
CLASSLESS ADDRESSING

- With the growth of the Internet, it was clear that a larger address space was needed as a long-term solution.
- Although the long-range solution has already been devised and is called IPv6, a short-term solution was to use the same address space but to change the distribution of addresses to provide a fair share to each organization.
- The short-term solution still uses IPv4 addresses, but it is called *classless addressing*.

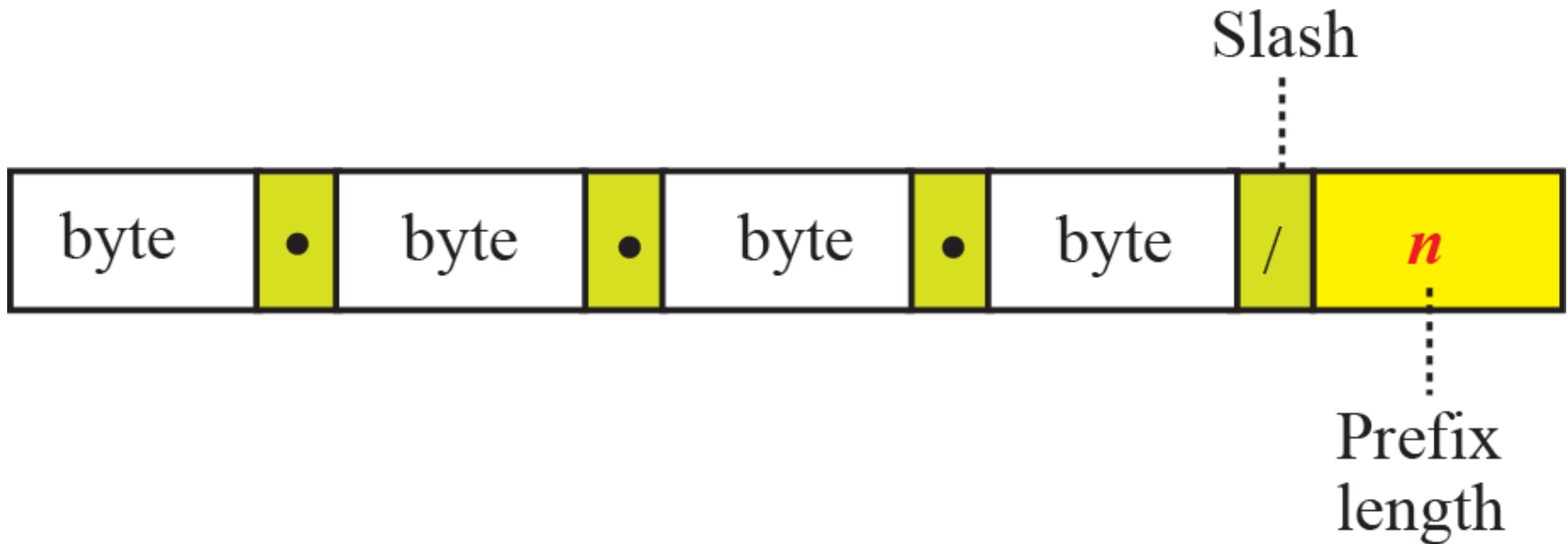
Variable-length blocks in classless addressing



In classless addressing, the prefix defines the network and the suffix defines the host.



The prefix length in classless addressing can be 1 to 32.



In classless addressing, we need to know one of the addresses in the block and the prefix length to define the block.

Example

In classless addressing, an address may not define the block the address belongs to. For example, the address 230.8.24.56 can belong to many blocks some of them are shown below with the value of the prefix associated with that block:

Prefix length:16	→	Block:	230.8.0.0	to	230.8.255.255
Prefix length:20	→	Block:	230.8.16.0	to	230.8.31.255
Prefix length:26	→	Block:	230.8.24.0	to	230.8.24.63
Prefix length:27	→	Block:	230.8.24.32	to	230.8.24.63
Prefix length:29	→	Block:	230.8.24.56	to	230.8.24.63
Prefix length:31	→	Block:	230.8.24.56	to	230.8.24.57

Example

The following addresses are defined using slash notations.

- a.** In the address 12.23.24.78/8, the network mask is 255.0.0.0. The mask has eight 1s and twenty-four 0s. The prefix length is 8; the suffix length is 24.
- b.** In the address 130.11.232.156/16, the network mask is 255.255.0.0. The mask has sixteen 1s and sixteen 0s. The prefix length is 16; the suffix length is 16.
- c.** In the address 167.199.170.82/27, the network mask is 255.255.255.224. The mask has twenty-seven 1s and five 0s. The prefix length is 27; the suffix length is 5.

Example 5 min. exercise

One of the addresses in a block is 167.199.170.82/27. Find the number of addresses in the network, the first address, and the last address.

Example 5 min. exercise

One of the addresses in a block is 167.199.170.82/27. Find the number of addresses in the network, the first address, and the last address.

Solution

The value of n is 27. The network mask has twenty-seven 1s and five 0s. It is 255.255.255.240.

- a. The number of addresses in the network is $2^{32 - n} = 32$.
- b. We use the AND operation to find the first address (network address). The first address is 167.199.170.64/27.

Address in binary:	10100111	11000111	10101010	01010010
Network mask:	11111111	11111111	11111111	11100000
First address:	10100111	11000111	10101010	01000000

Example 5 min. exercise

- c.** To find the last address, we first find the complement of the network mask and then OR it with the given address:
The last address is 167.199.170.95/27.

Address in binary:	10100111	11000111	10101010	01010010
Complement of network mask:	00000000	00000000	00000000	00011111
Last address:	10100111	11000111	10101010	01011111

Example – Address admin

An ISP has requested a block of 1000 addresses. The following block is granted.

- a.** Since 1000 is not a power of 2, 1024 addresses are granted ($1024 = 2^{10}$).
- b.** The prefix length for the block is calculated as $n = 32 - 10 = 22$.
- c.** The beginning address is chosen as 18.14.12.0 (which is divisible by 1024).

The granted block is 18.14.12.0/22. The first address is 18.14.12.0/22 and the last address is 18.14.15.255/22.

Example

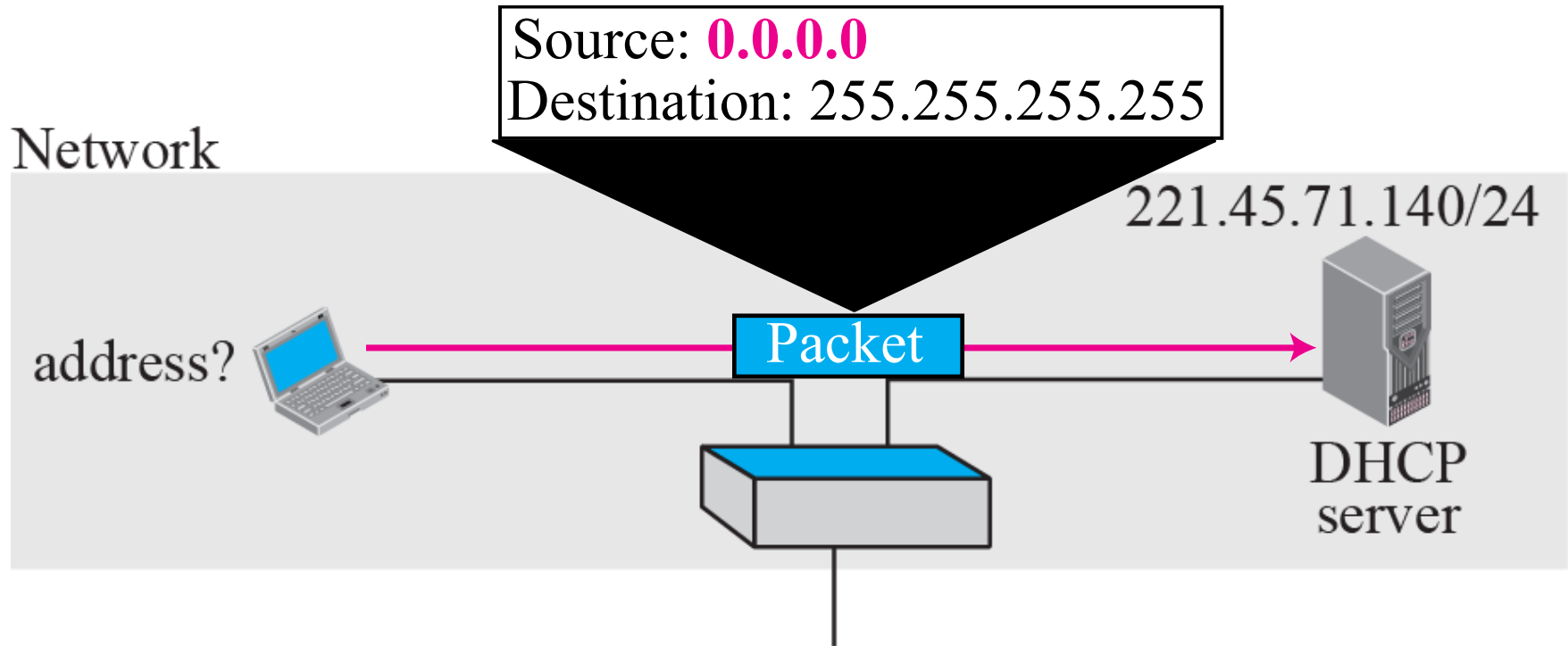
Assume your organization has been given a class A block as 73.0.0.0 in the past.

If the block is not revoked by the authority, the classless architecture assumes that the organization has a block 73.0.0.0/8 in classless addressing.

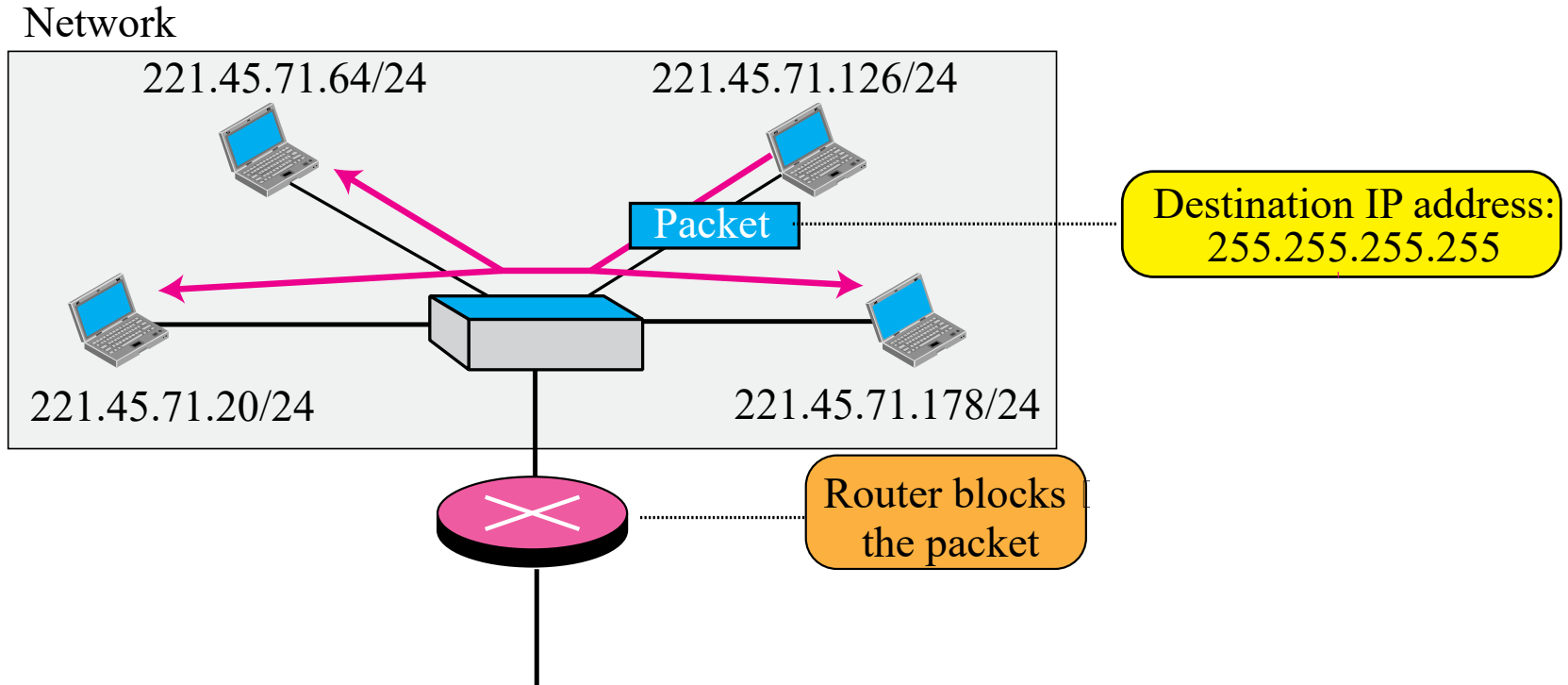
SPECIAL ADDRESSES

- In classful addressing some addresses were reserved for special purposes.
- The classless addressing scheme inherits some of these special addresses from classful addressing.

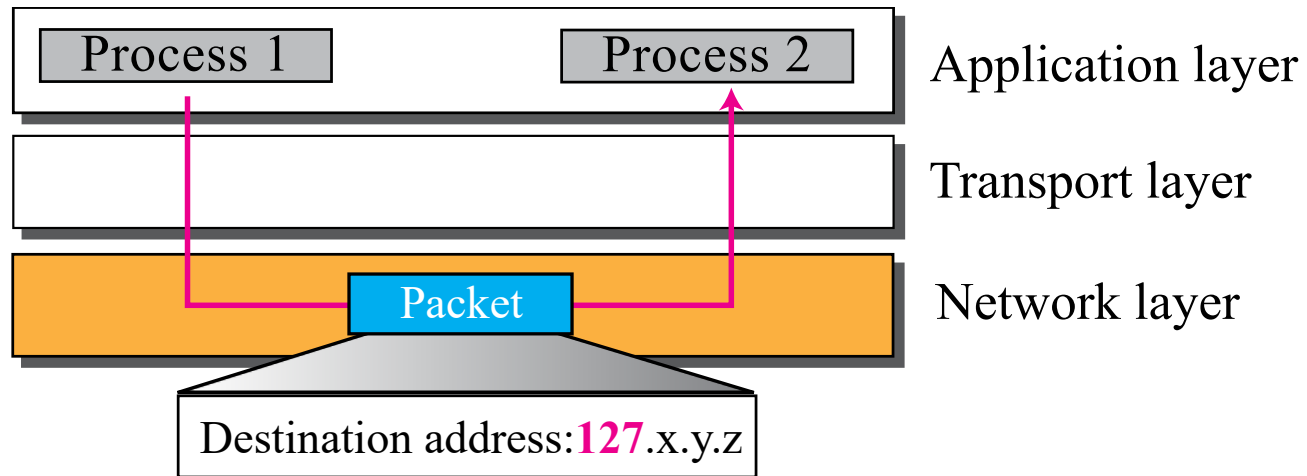
Example of using the all-zero address



Example of limited broadcast address



Example of loopback address



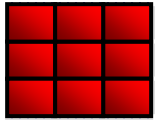
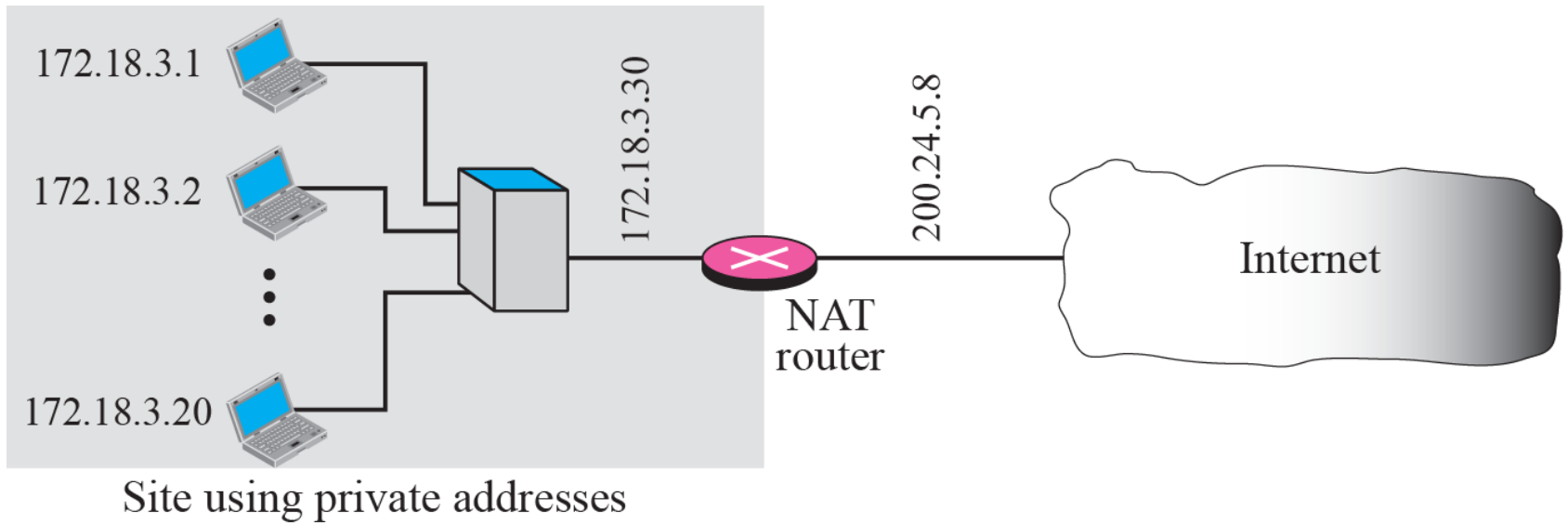


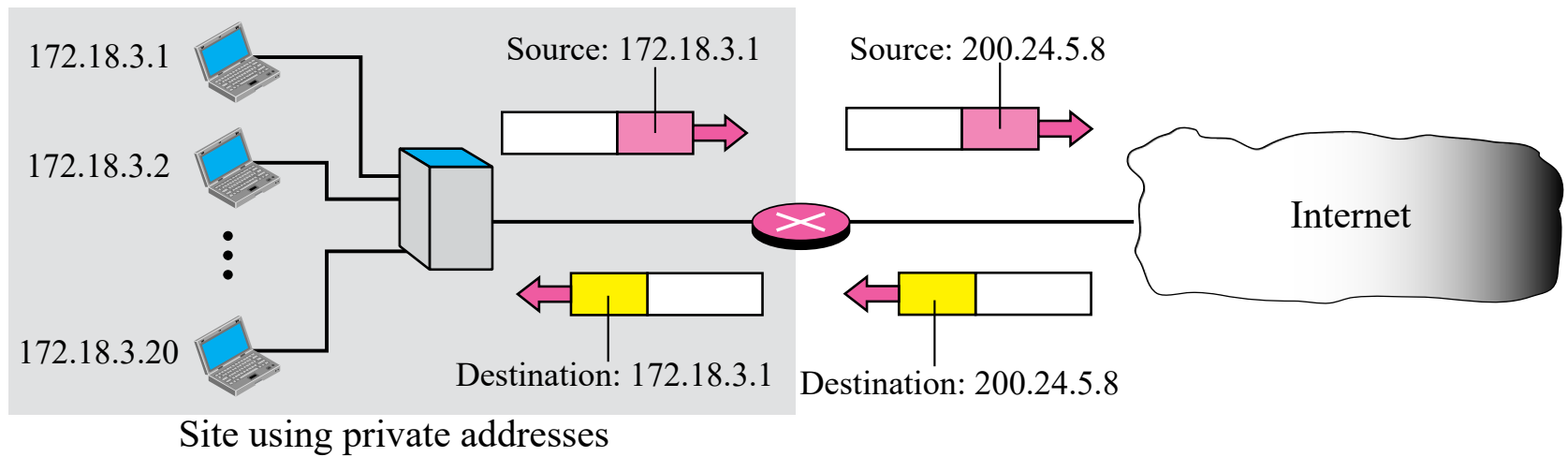
Table 5.2 *Addresses for private networks*

<i>Block</i>	<i>Number of addresses</i>	<i>Block</i>	<i>Number of addresses</i>
10.0.0.0/8	16,777,216	192.168.0.0/16	65,536
172.16.0.0/12	1,047,584	169.254.0.0/16	65,536

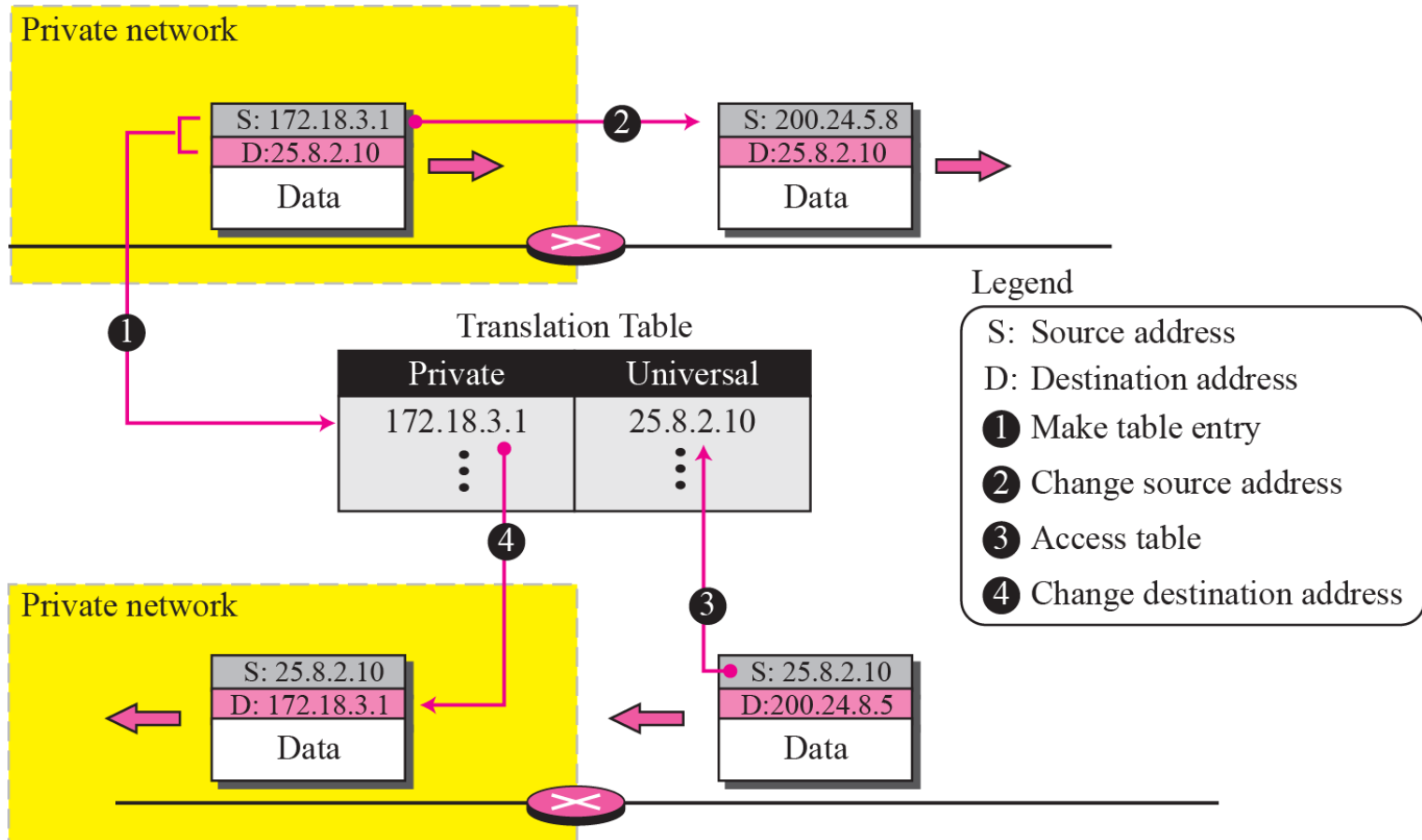
- If the business grows, the ISP may not be able to grant the demand because the addresses before and after the range may have already been allocated to other networks.
- In most situations only a portion of computers in a small network need access to the Internet simultaneously.
- A technology that can help in this cases is *network address translation (NAT)*.



Address resolution



Translation



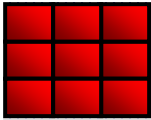


Table 5.3 *Five-column translation table*

<i>Private Address</i>	<i>Private Port</i>	<i>External Address</i>	<i>External Port</i>	<i>Transport Protocol</i>
172.18.3.1	1400	25.8.3.2	80	TCP
172.18.3.2	1401	25.8.3.2	80	TCP
...	...	...	...	...

References

**TCP/IP Protocol Suite (4th Edition),
by Behrouz A. Forouzan – Used the text's slide deck**

End of session